



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ARS AGENCY - UTHANGARAI

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

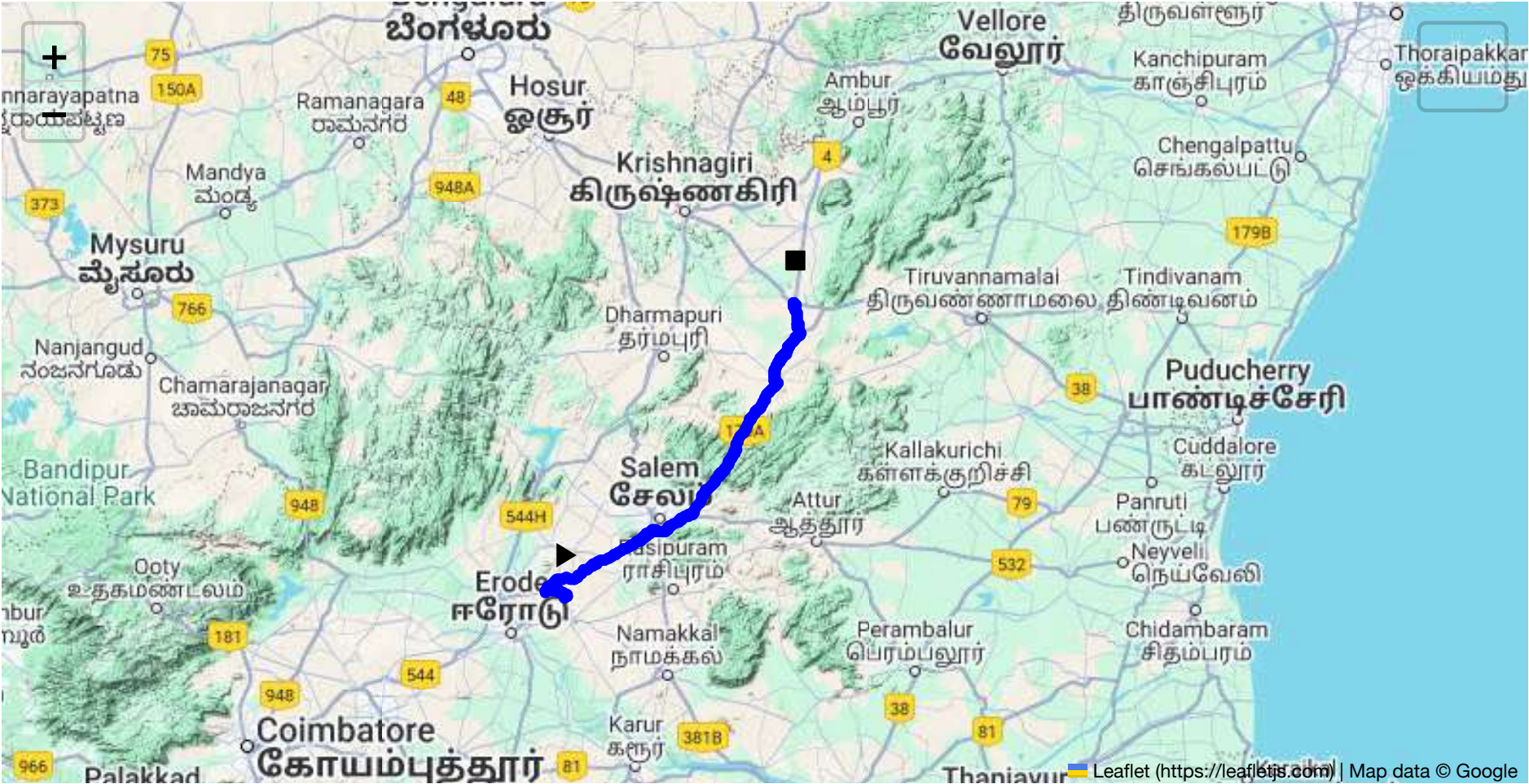
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 147.44 km
Estimated Duration: 3.4 hours
Adjusted Duration (Heavy Vehicle): 4.2 hours

Start: (11.4381, 77.8734)
End: (12.270948, 78.528177)



Welcome to the Journey Risk Management Study

Here is a detailed analysis of the route from Sangagiri to Uthangarai, focusing on identifying potential hazards and preparing for transportation of hazardous materials:

- Overview of the Route Map:** The route spans approximately 147.44 kilometers and involves traveling through various highways and regional roads. Key segments include Bhavani Main Road, Nagapattinam - Coimbatore - Gundlupet Highway, and Dharmapuri - Harur Road. The route passes through several towns and medium-sized cities, implicating varying traffic conditions and infrastructure quality.
- Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu generally experiences a tropical climate, with high temperatures year-round. The region can face heavy rainfall during the monsoon from October to December, which may result in water-logging and slippery roads. Fog can also be a concern during early morning hours, affecting visibility.
- Analysis of Traffic Patterns:** The areas around Sangagiri, Salem, and Dharmapuri usually experience higher traffic volumes. Peak hours are typically morning (8:00 to 10:00 AM) and evening (5:00 to 7:00 PM), where congestion may occur, particularly in city centers and near schools. Traffic in Salem is notably heavier due to it being a major junction point.
- Assessment of Road Quality and Infrastructure:** The Nagapattinam - Coimbatore - Gundlupet Highway is generally well-maintained, but some stretches near local roads may have uneven surfaces and potholes. Road quality in rural areas can vary, demanding cautious driving and awareness of sudden rough patches or narrow roads.
- Suggestions for Alternative Routes for Emergencies:** In case of an emergency, an alternative route includes diverting via NH44 that connects Salem to Dharmapuri and then accessing local roads to reach Uthangarai. This would add distance but can serve as a bypass during unexpected roadblocks or severe weather conditions.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Transport of hazardous materials is regulated, requiring adherence to safety standards such as proper labeling, documentation, and vehicle compliance with safety features. Time restrictions may apply, usually limiting travel through densely populated areas during peak hours or night.
7. **Overview of Historical Incidents:** There have been reports of occasional accidents involving heavy vehicles due to speeding and poor road visibility, particularly during adverse weather conditions. However, specific data on hazardous materials incidents is limited in public records.
8. **Environmental Considerations and Sensitive Areas:** The route passes near agricultural lands; thus, pollution control is essential. Preventing leaks, careful waste management, and swift clean-up measures to avoid contaminating soil and water resources are imperative.
9. **Analysis of Communication Coverage:** Most of the route enjoys good mobile network coverage, but there may be potential dead zones, particularly in rural or forested areas between Dharmapuri and Harur. It is advisable to have offline navigation tools available.
10. **Estimated Emergency Response Times:** Emergency services in urban areas like Salem and Dharmapuri typically respond within 30 minutes to an hour. In more rural segments, response times may extend to 1-2 hours due to infrastructure and distance challenges.
11. **Overall Summary of Risk Assessment:** The transportation route presents standard risks associated with monsoon-related weather, traffic congestion, and infrastructure variability. Prior planning for peak traffic, adherence to local regulations, and emergency preparedness are crucial. Regular communication checks and having alternative routes can mitigate delays and enhance safety.

These insights cater to truck drivers transporting hazardous materials, ensuring safety on the road while complying with regulations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Roundabout	High	12.26577, 78.53533	15 KM/Hr
2	U-Turn	High	11.6180669, 78.1109048	10 KM/Hr
3	U-Turn	High	11.6128439, 78.1059968	10 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	High	11.44900, 77.87406	15 KM/Hr
7	Turn	High	11.45353, 77.85697	15 KM/Hr
8	Turn	Medium	11.45781, 77.85203	30 KM/Hr
9	Turn	High	11.46307, 77.84941	15 KM/Hr
10	Turn	Medium	11.45546, 77.81800	30 KM/Hr
11	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
12	Turn	Medium	11.53578, 77.95802	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
14	Turn	High	11.62106, 78.10767	15 KM/Hr
15	Blind Spot	Blind Spot	11.61217, 78.10439	10 KM/Hr
16	Blind Spot	Blind Spot	11.61801, 78.11095	10 KM/Hr
17	Blind Spot	Blind Spot	11.61285, 78.10608	10 KM/Hr
18	Turn	High	11.63306, 78.12728	15 KM/Hr
19	Turn	Medium	11.62861, 78.16526	30 KM/Hr
20	Turn	Medium	11.65786, 78.20436	30 KM/Hr
21	Turn	High	11.66388, 78.20672	15 KM/Hr
22	Turn	Medium	11.67129, 78.22978	30 KM/Hr
23	Turn	Medium	11.67116, 78.23102	30 KM/Hr
24	Turn	High	11.67145, 78.23141	15 KM/Hr
25	Turn	High	11.67198, 78.23148	15 KM/Hr
26	Turn	Medium	11.67440, 78.23914	30 KM/Hr
27	Turn	Medium	11.81235, 78.34084	30 KM/Hr
28	Turn	Medium	11.99554, 78.44553	30 KM/Hr
29	Turn	Medium	12.01577, 78.45673	30 KM/Hr
30	Turn	Medium	12.04335, 78.48368	30 KM/Hr
31	Turn	Medium	12.04504, 78.48335	30 KM/Hr
32	Turn	High	12.04990, 78.47475	15 KM/Hr
33	Turn	Medium	12.06023, 78.47792	30 KM/Hr
34	Turn	Medium	12.06932, 78.47655	30 KM/Hr
35	Turn	Medium	12.07112, 78.47724	30 KM/Hr
36	Turn	Medium	12.11669, 78.49766	30 KM/Hr
37	Turn	Medium	12.17764, 78.54800	30 KM/Hr
38	Turn	Medium	12.17847, 78.54856	30 KM/Hr
39	Turn	Medium	12.18223, 78.54843	30 KM/Hr
40	Turn	High	12.18614, 78.54962	15 KM/Hr
41	Turn	Medium	12.20186, 78.54375	30 KM/Hr
42	Turn	Medium	12.20953, 78.54358	30 KM/Hr
43	Turn	High	12.21030, 78.54402	15 KM/Hr
44	Turn	Medium	12.25956, 78.53301	30 KM/Hr
45	Turn	Medium	12.26092, 78.53324	30 KM/Hr
46	Turn	Medium	12.26207, 78.53436	30 KM/Hr
47	Turn	High	12.26689, 78.53108	15 KM/Hr
48	Turn	Medium	12.27021, 78.52951	30 KM/Hr
49	Turn	Medium	12.27100, 78.52821	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
4	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
6	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
11	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
12	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
15	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
16	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
17	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
18	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
19	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
20	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
23	hospital	Dr.Thiru Neuro Multi Speciality	11.6623376, 78.203353	30 km/h	Medium
25	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
26	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
27	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
33	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium
34	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
35	hospital	Government Hospital, Uthangarai	12.26612, 78.535949	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
5	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
9	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
10	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
21	school	St.joseph's matric hr.sec school	11.6419382, 78.1709055	30 km/h	Medium
22	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
24	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
28	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
29	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
30	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
31	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium
32	school	Government Girls Hr Sec School	12.0488709, 78.4762447	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.26577, 78.53533



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.6180669, 78.1109048



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.6128439, 78.1059968



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



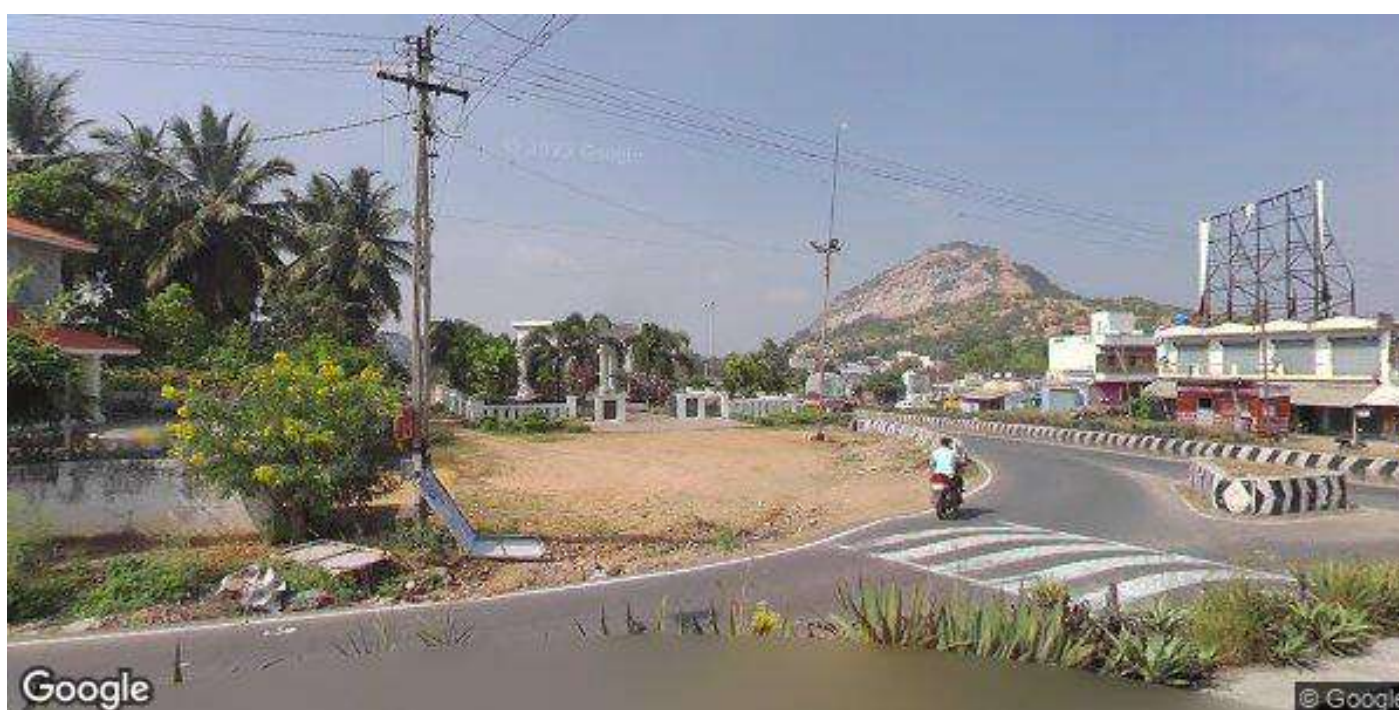
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53578, 77.95802



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62106, 78.10767



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61217, 78.10439



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61801, 78.11095



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61285, 78.10608



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63306, 78.12728



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62861, 78.16526



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65786, 78.20436



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66388, 78.20672



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67129, 78.22978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67116, 78.23102



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67145, 78.23141



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67198, 78.23148



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67440, 78.23914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81235, 78.34084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99554, 78.44553



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01577, 78.45673



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Speed Limit: 30 KM/Hr
Coordinates: 12.04335, 78.48368



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Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04504, 78.48335



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.04990, 78.47475



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.06023, 78.47792



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Speed Limit: 30 KM/Hr
Coordinates: 12.06932, 78.47655



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Speed Limit: 30 KM/Hr
Coordinates: 12.07112, 78.47724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11669, 78.49766



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17764, 78.54800



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17847, 78.54856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.18223, 78.54843



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.18614, 78.54962



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20186, 78.54375



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20953, 78.54358



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21030, 78.54402



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25956, 78.53301



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26092, 78.53324



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26207, 78.53436



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.26689, 78.53108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.27021, 78.52951



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.27100, 78.52821
